higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N6

(8080376)

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09:00–12:00

Nonprogrammable calculators may be used.

This question paper consists of 8 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
DIGITAL ELECTRONICS N6
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1: COMPUTER SYSTEMS

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.4) in the ANSWER BOOK.

1.1.1 A device that takes a single input and distributes it over several outputs:

A Modem
B Multiplexer
C Demultiplexer
D Analogue-to-digital converter

1.1.2 A transducer that experiences a decrease in resistance when the temperature increases:

A PTC-thermistor
B NTC-thermistor
C Thermocouple
D Thermometer

1.1.3 The period between application of an input and the output reaching a specified point of its final value:

A Resolution
B Conversion time
C Settling time
D Quantising error

1.1.4 The maximum output voltage of a DAC or an ADC:

A Step size
B Threshold voltage
C Full-scale voltage
D Analogue voltage

(4 × 1) (4)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answers and write only 'True' or 'False' next to the question number (1.2.1–1.2.4) in the ANSWER BOOK.

1.2.1 A thermocouple works because of the Seebeck effect.

1.2.2 The size of the instruction repertoire is governed by the size of the op-code.

1.2.3 BASIC is a low-level language.

1.2.4 A feasibility study is conducted by a systems analyst.

(4 × 1) (4)

1.3 Choose an item from COLUMN B that matches a description in COLUMN A. Write only the letter (A–F) next to the question number (1.3.1–1.3.4) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.3.1	Error in a program	A	ROM
1.3.2	Obsolete microprocessor	B	RAM
1.3.3	Graphical representation of a program	C	Pentium
		D	flow chart
1.3.4	Volatile memory	E	bug
		F	8085

(4 × 1) (4)

1.4 A team of experts led by a system analyst must compile a nontechnical report on the feasibility of whether or not a system can be automated.

List SIX criteria that this feasibility study must be based on. (6)

1.5 Sketch a simple graph to illustrate what is meant by a quantisation error in an analogue-to-digital converter. (2)

[20]

QUESTION 2: TRANSMISSION, DATA ACQUISITION AND RELATED HARDWARE

2.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (2.1.1–2.1.4) in the ANSWER BOOK.

2.1.1 The interrupt generated by an event outside of the computer:

A Priority
B Internal
C Nonmaskable
D External

2.1.2 The port into which a dongle-type modem is inserted:

A USART
B UART
C USB
D I/O

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2.1.3 Which communication system requires no start and stop bits?

A Frequency-shift keying
B Wireless transmission
C Synchronous transmission
D Asynchronous transmission

2.1.4 If a 9-bit ADC has a threshold voltage of 1 mV, what will its full-scale voltage be?

A 2,047 V
B 1,023 V
C 0,511 V
D 0,256 V

(4 × 1) (4)

2.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (2.2.1–2.2.4) in the ANSWER BOOK.

2.2.1 FSK tones are not audible.

2.2.2 A subroutine can be considered as a type of interrupt.

2.2.3 PCM stands for pulse-coded modulation.

2.2.4 Hamming code is commonly used as an error-detecting code.

(4 × 1) (4)

2.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–F) next to the question number (2.3.1–2.3.4) in the ANSWER BOOK.

COLUMN A		COLUMN B	
2.3.1	Used in up/down counters	A	Hamming code
2.3.2	Used in error detection	B	Gray code
2.3.3	Source of transmission noise	C	handshaking
2.3.4	Process of synchronising modems	D	solar flare
		E	XOR gate
		F	XNOR gate

(4 × 1) (4)

2.4 Show by means of a simple diagram how frequency-shift keying (FSK) works. (3)

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2.5 Aside from using a demultiplexer (demux) for its primary function of returning multiplexed signals to their original format, what else can a demux be used for? (1)

2.6 A PCM system has to transmit the following packages using NRZ (nonreturn to zero):

1110 1000 0011 1111

Graphically show what the transmission would look like by using ALL FOUR bits in the string. (4)
[20]

QUESTION 3: COMPUTER ARCHITECTURE

Control units vary from one computer to another. Base your answers to the following questions on Von Neumann architecture where a common RAM holds ALL data and program instructions, necessitating a fetch and an execute routine.

3.1 Draw a neat, fully labelled block diagram of a control unit of a Von Neumann computer. The micro-instruction unit is a 6-bit ring counter. The system is synchronous and each microprogram has SIX micro-instructions, making 'do nothing' phases possible. Show ALL gates and interconnections between the micro-instruction unit, the program counter, the RAM, the instruction unit, the decoder and the arithmetic unit. (10)

3.2 Use the following table to show what happens after each pulse from the micro-instruction unit in QUESTION 3.1. Clearly separate each micro-instruction from the next and show how the micro-instructions are grouped into routines. Use block diagrams or descriptions or both to clearly show what happens when the control unit receives the instruction to subtract a number from the accumulator.

INSTRUCTION LOCATION	INSTRUCTION WORD	DESCRIPTION
0001	1001 0010	The content of address 0010 is 0111 0111. This must be subtracted from the content of the accumulator.

(10)
[20]

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QUESTION 4: HIGH-LEVEL PROGRAMMING

4.1 Consider the following FORTRAN program segment:

```
      PINK = 8
      FLOYD = 8
355 RESULT = PINK + FLOYD
      FLOYD = FLOYD + 7
      IF (RESULT.LT.35) THEN 355
      PRINT, PINK
      PRINT, FLOYD
      PRINT, RESULT
      STOP
      END
```

Draw a table with the following headings: PASS, PINK, FLOYD and RESULT. Now clearly show how each value changes with each successive pass. Finally show, beneath the table, what the final printout will look like. (10)

4.2 The following is a schematic drawing of a stack with its contents:

Stack pointer
18

Stack:

Address	Contents
16	04_{16}
17	23_{16}
18	$F1_{16}$

If the number $2B_{16}$ is pushed onto the stack, redraw the stack and the stack pointer to show what the contents would look like now. (3)

4.3 Give ONE everyday example where batch processing would be necessary. (1)

4.4 The following is a section of a program in machine code:

INSTRUCTION	OP-CODE	ADDRESS	DESCRIPTION
0000	0011	1001	LOAD ACCUMULATOR
0001	0111	1010	ADD
0010	0111	1101	ADD
0011	0110	1100	SUB
0100	0111	1011	SUB

The following are the address contents to which the above program refers:

ADDRESS	ADDRESS CONTENTS
1001	0011 0011
1010	0100 0100
1011	0001 0001
1100	0111 0111
1101	1000 1000

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Rewrite each instruction number one below the other and clearly show next to it the accumulator contents after each instruction has been executed. (5)

4.5 What is the lowest-level language (the code that ALL digital devices ultimately operate in)? (1)

[20]

QUESTION 5: NUMBER SYSTEMS

5.1 The following word is received in Hamming code:

$1010111011_{hamming}$

Find the fault in the word by showing ALL steps involved. Clearly state the bit number on which the fault lies and then rewrite the correct word clearly indicating which bit has been corrected. (10)

5.2 Write the following floating point number in decimal showing ALL steps:

0 000 0 101000000 (3)

5.3 Rewrite the following 2.4.2.1 code in 8.4.2.1 code:

$1011\ 1110\ 0100_{2.4.2.1}$ (3)

5.4 Rewrite the following binary number in Gray code:

1111001_2 (2)

5.5 Write down the general rule for De Morgan's theorem using variables A, B, C and D. (2)

[20]

TOTAL: 100

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